CS 249 – Assignment: VM Resource Monitoring and Benchmarking

Name: Mina Alvarez | Student ID: 017904118

Generated sample submission with dummy data, tables, and figures for PDF extraction evaluation.

This sample report summarizes a synthetic benchmarking exercise on a Linux virtual machine. The write-up includes console screenshots, generated charts, and summary tables to emulate a realistic systems assignment.

1. Test Environment

- Ubuntu 22.04 VM with 4 vCPUs and 8 GB RAM
- Python service exposed on localhost:8080
- Load driver issuing mixed read/write requests for 15 minutes
- Monitoring via htop, vmstat, and custom latency logs

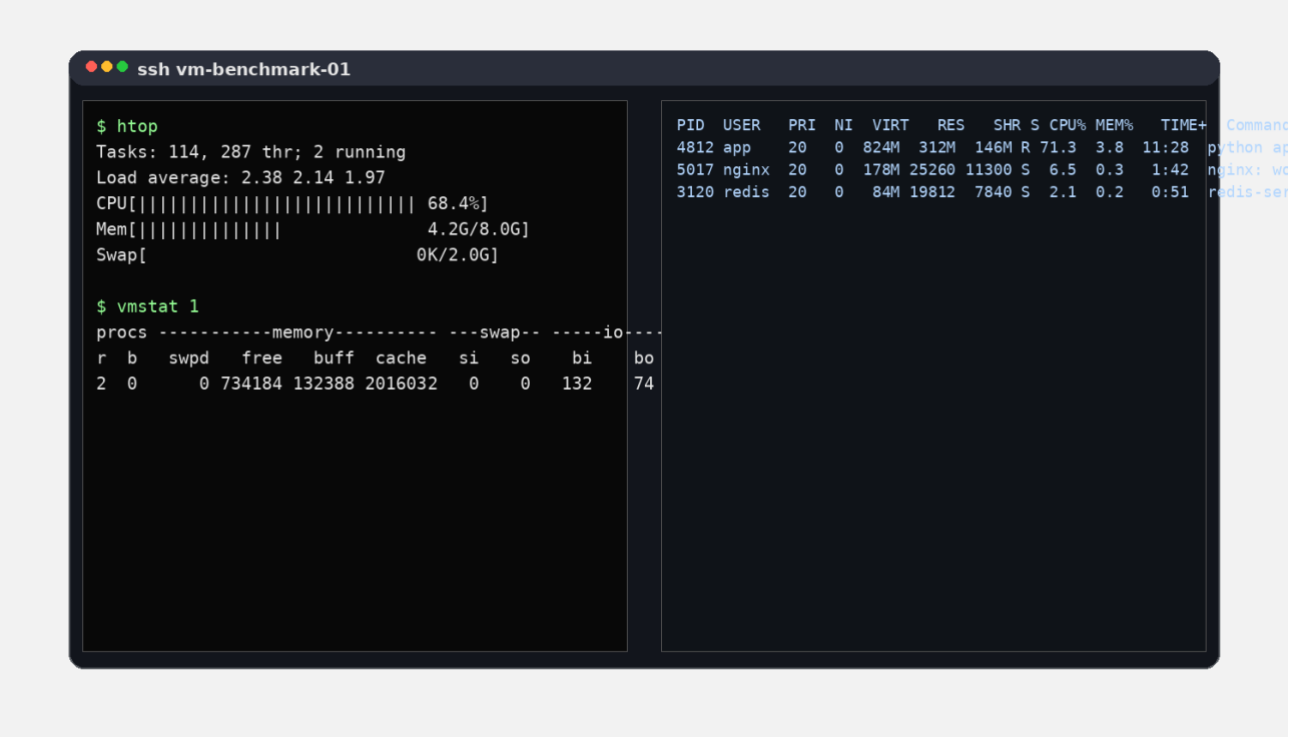

Figure 1. Mock terminal output during the load test.

2. Benchmark Figures

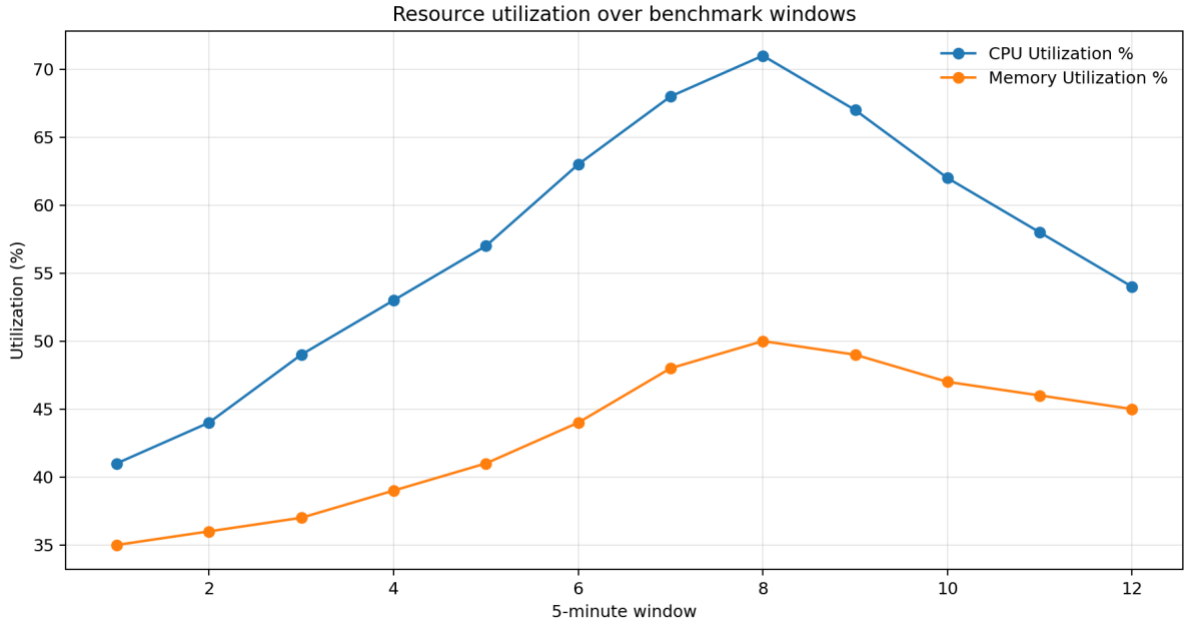

Figure 2. CPU and memory utilization observed across the benchmark.

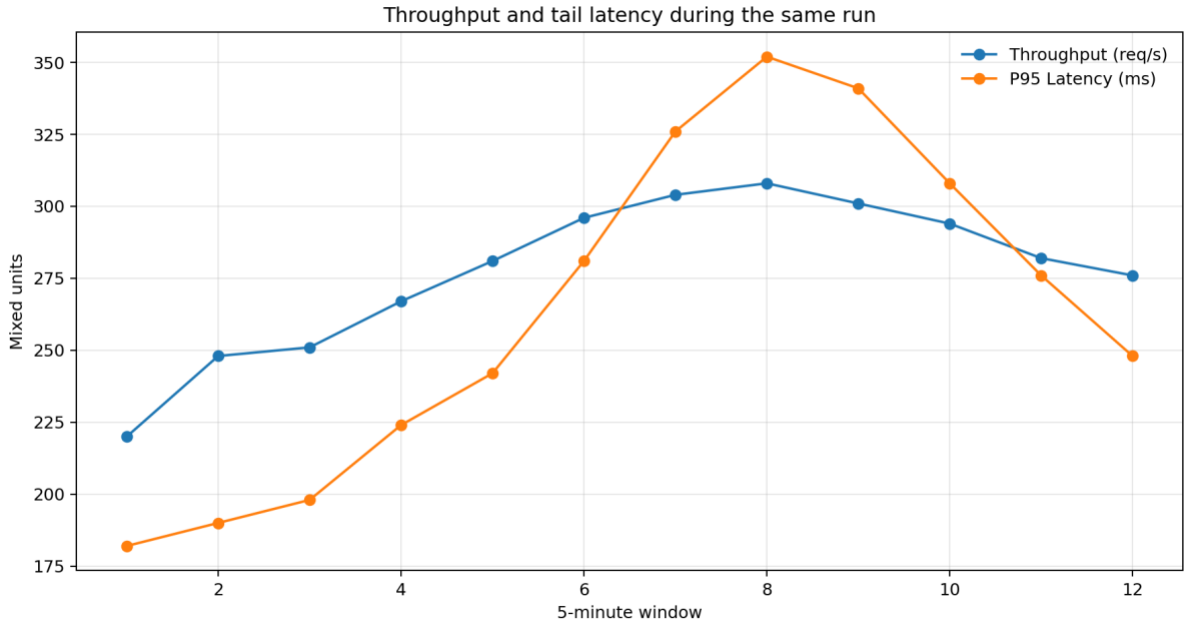

Figure 3. Throughput and tail-latency progression across the test.

3. Summary Table and Findings

Run	RPS	Avg Latency (ms)	P95 (ms)	CPU Peak %	Outcome
baseline	228	146	212	54	stable
cache enabled	281	132	198	57	stable
write burst	244	188	326	71	watch
reduced worker count	208	201	352	68	degraded

1. The cache-enabled scenario produced the highest throughput without exhausting CPU or memory headroom.
2. Tail latency increased sharply during the write-burst phase, which is consistent with queue buildup inside the Python service.
3. Reducing worker count lowered resource usage slightly but pushed response times beyond the acceptable threshold.
4. A mix of console screenshots, charts, and compact summary tables makes this document useful for PDF extraction testing.